

Aussie Primary Academy | Australian Curriculum v9.0 | Learn, Investigate and Explain

Name: _____

Date: _____

LEARNING INTENTION	describe properties of solids and liquids and investigate how adding or removing heat leads to changes of state
KEY IDEA	Solids keep a shape; liquids flow and keep volume while taking the container's shape. Heating can cause melting and cooling can cause freezing; the material remains the same substance.
WORKED MODEL	Ice melts to liquid water when it gains heat; liquid water freezes when it loses heat.

SUCCESS CRITERIA - I CAN

[] use key vocabulary [] classify or model accurately [] explain using evidence [] correct a misconception

A. BUILD THE SCIENCE IDEA

1. Define and use these terms: solid and liquid.

2. Match examples to: solid, liquid, melting, freezing.

3. Annotate the worked model with two observations and one scientific explanation.

B. APPLY AND EXPLAIN

4. Sort six examples from this context: ice, water, wax, honey, a wooden block and cooking oil

5. Explain two classifications using observable evidence, not appearance alone.

6. Identify the changed condition, the effect and one condition that should stay the same.

C. INDEPENDENT CHECK

7. Draw or construct a labelled scientific model that shows the relationship.

8. Write a three-sentence explanation using because, therefore and evidence.

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D. APPLY SCIENCE IN CONTEXT

9. Case study: Students place equal ice cubes in a sunny place, shade and an insulated container and record melting time.

10. Record the important observations in a two-column evidence table.

11. Write a claim-evidence-reasoning conclusion and state one limitation.

E. REASON, EVALUATE AND TRANSFER

12. Find and correct the inaccurate statement in the case study; explain why it is inaccurate.

13. Apply the same science idea to a different everyday situation and predict what would happen.

F. STUDENT REVIEW - SHOW WHAT YOU UNDERSTAND

[] I can describe properties of solids and liquids and investigate how adding or removing heat leads to changes of state.

Circle one: independently | with one reminder | with support

Evidence from my work that proves I understand: _____

The most difficult science idea was _____ because _____

One correction or improvement I made: _____

A question I still have: _____

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Accept equivalent responses when the student uses accurate science ideas, relevant evidence and clear reasoning.

COMPLETE ANSWER KEY

1. Definitions should mention shape, flow and state change.
2. Ice/wax/block are solids; water/honey/oil liquids at room conditions.
3. Adding heat causes melting; removing heat causes freezing.
4. Sort by observed behaviour, noting temperature conditions.
5. Use flow and fixed shape as evidence.
6. Changed: location/container; measured: melting time; control cube size and starting state.
- 7-8. Particle-free state model or process arrows accepted at this level.
9. Faster melting indicates greater heat transfer under test conditions.
10. Table records condition and melting observations/time.
11. Sun/shade temperature and only one cube may limit conclusion.
12. Melting does not create a new substance.
13. Accept chocolate, butter or water example with accurate heat direction.

TEACHER CHECK AND NEXT STEP

[] Uses accurate vocabulary and a complete cause-and-effect explanation.

[] Supports conclusions with observations or data.

Next step: revisit one misconception, then transfer the idea to a new example.

COMMON MISCONCEPTION / WATCH OUT

Liquids are not defined by being watery; thick honey is still a liquid.

Melting and dissolving are different processes.

EXTENSION AND RE-TEACH

Re-teach: model one difficult item, complete one together, then ask the student to explain an independent correction.

Extension: apply the science idea to a new real-world example and justify the explanation with observations or data.